COMPETENCY STANDARD
FOR
QUALITY CONTROL IN YARN MANUFACTURING
(TEXTILE SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance
Government of the People's Republic of Bangladesh

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The Competency Standard for **Quality Control in Yarn Manufacturing** is designed to guide the development of curricula, teaching materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry, so that individuals who pass through the set standard via assessment would be qualified and settled for a relevant job.
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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades, which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The textile sector is the largest manufacturing and export-earning sector of Bangladesh, and yarn manufacturing (spinning) is the primary value-adding stage of the textile value chain. In a spinning mill, however, installed capacity alone does not determine competitiveness. Profitability and market acceptance depend on how efficiently the machines and manpower are utilized, how much of the raw material is converted into saleable yarn, and how consistently the yarn meets the quality characteristics demanded by weaving, knitting, dyeing and export buyers. A small improvement in machine efficiency, waste percentage or yarn realization, or a small reduction in yarn faults and contamination, translates directly into significant cost savings and higher value realization for the mill.

Bangladeshi spinning mills continue to report avoidable losses arising from high end breakage rates, excessive waste, unplanned downtime, contamination, count variation and customer complaints. These losses persist largely because there is a shortage of workers trained specifically to measure, analyse and improve productivity and quality using structured techniques and standard test methods. Existing training programs concentrate mainly on machine operation, while the analytical and laboratory competencies required for process control, testing, waste control, statistical process control and continuous improvement have remained largely un-standardized.

This Competency Standard has therefore been developed to prepare a competent workforce for productivity improvement and quality control in yarn manufacturing. It covers assessment of raw material quality, in-process quality control from mixing to roving, yarn testing and evaluation, identification and control of yarn faults and contamination, measurement and improvement of machine and manpower productivity, waste control and cost reduction, application of quality management systems and continuous improvement tools, and monitoring of the process environment, machine condition and instrument calibration.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and to ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competency and its corresponding Elements.

OVERVIEW:

A Competency Standard is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

1. provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
2. enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies.
3. encourages the development and delivery of flexible training that suits individual and industry requirements.
4. encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standards are developed by a working group comprised of national and international process experts and the participation of experts from the industry to identify the competencies required of an occupation in a particular sector.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at a training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

1. Reference to Industry Sector, Occupational Title and Occupational Description
2. Unit code
3. Unit title
4. Unit descriptor
5. Unit of Competency
6. Elements and performance criteria
7. Variables and range statement
8. Evidence guides

Together all the parts of a Unit of Competency:

1. Describe a work activity
2. Guide the assessor in determining whether the candidate is competent.

This Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competency:

1. An overview of all Units of Competency for the occupation and their corresponding duration required for completion of training.
2. The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

UNITS & ELEMENTS AT A GLANCE:

Generic Competencies (20 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TEX-QCYM-01-G	Apply Occupational Health and Safety (OHS) Practices in the Workplace	1. Identify OHS policies and procedures 2. Apply personal health and safety practices 3. Report hazards and risks 4. Respond to emergencies	10
SICIP-TEX-QCYM-02-G	Work in a Team Environment	1. Identify team goals and processes 2. Communicate and cooperate with team members 3. Work as a team member 4. Solve problems as a team member	10
		Total Hours	**20 Hrs.**

Sector Specific Competencies (20 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TEX-QCYM-01-S	Work in the Textile Sector	1. Identify the textile sector and its main activities 2. Understand basic roles in textile production processes 3. Recognize common tools and equipment used 4. Understand simple quality requirements in textile work	10
SICIP-TEX-QCYM-02-S	Apply Green Practices in the Textile Sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
		Total Hours	**20 Hrs.**

Occupation Specific Competencies (320 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TEX-QCYM-01-O	Interpret Quality Concepts in Yarn Manufacturing	1. Interpret quality concepts and standards. 2. Interpret yarn testing methods. 3. Identify factors affecting quality control procedures. 4. Interpret quality management system.	50
SICIP-TEX-QCYM-02-O	Perform Fibre and Raw Material Quality Assessment	1. Prepare for fibre and raw material testing 2. Collect and condition samples 3. Test fibre properties 4. Interpret fibre test results 5. Record and report fibre quality data	60
SICIP-TEX-QCYM-03-O	Perform In-process Quality Control in Spinning Preparatory Processes	1. Prepare for in-process quality control 2. Check blow room, lap and chute feed quality 3. Check card sliver quality 4. Check draw frame, comber and roving quality 5. Record and report in-process quality data	90
SICIP-TEX-QCYM-04-O	Perform Yarn Quality Testing and Evaluation	1. Prepare the laboratory and instruments for yarn testing 2. Test yarn count, twist and moisture regain 3. Test yarn strength and elongation 4. Test yarn evenness, imperfections and hairiness 5. Interpret, record and report yarn test results	70
SICIP-TEX-QCYM-05-O	Identify and Control Yarn Faults and Contamination	1. Identify yarn, package and process faults 2. Analyse causes of faults 3. Control contamination and foreign fibre 4. Apply corrective and preventive actions	50
		Total Hours	**320 Hrs.**

COMPETENCY STANDARD: QUALITY CONTROL IN YARN MANUFACTURING

The Generic Competencies

Unit of Competency: **APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE**	**Nominal Duration:** 10 Hrs.	**Unit Code:** SICIP-TEX-QCYM-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply Occupational Health and Safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures; applying personal health and safety practices; reporting hazards and risks; and responding to emergencies.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1. 1 **OHS policies** and safe operating procedures are read and understood. 1. 2 Safety signs and symbols are identified and followed. 1. 3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Apply personal health and safety practices	2. 1 OHS policies and procedures are followed and practiced. 2. 2 **Personal Protective Equipment (PPE)** is selected and used. 2. 3 Personal hygiene is maintained.
3. Report hazards and risks	3.1 **Hazards and risks** are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to the authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to emergencies	4. 1 Alarms and warning devices are responded to. 4. 2 **Emergency response plans and procedures** are implemented. 4. 3 **First aid procedure** is applied during emergency situations.

Range of Variables:

Variable	Range (May include but not limited to)
1. OHS policies	1. 1 International OHS requirements 1. 2 Bangladesh standards for OHS

Variable	Range (May include but not limited to)
	1. 3 Bangladesh Labour Act and Rules 1. 4 Building Code 1. 5 Fire Safety Rules and Regulations 1. 6 Industry guidelines and mill safety manual
2. Personal Protective Equipment (PPE)	2. 1 Nose/face mask (dust mask) 2. 2 Ear plugs or ear muffs 2. 3 Hand gloves 2. 4 Safety shoes 2. 5 Head cap/helmet 2. 6 Apron and overalls 2. 7 Goggles and safety glasses 2. 8 Hair net
3. Hazards and risks	3.1 Physical hazards (moving machine parts, nip points, noise, vibration, heat) 3.2 Chemical hazards (lubricants, cleaning agents, solvents) 3.3 Biological hazards (bacteria, fungus in damp areas) 3.4 Ergonomic hazards (repetitive work, standing, manual lifting) 3.5 Electrical hazards 3.6 Fire hazards (cotton dust, fly accumulation, static electricity)
4. Emergency response plans and procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first aid 4.5 Electrical accident response 4.6 Emergency machine stop procedures
5. First aid procedure	5. 1 Washing of open wound 5. 2 Washing chemically infected area 5. 3 Applying bandage 5. 4 Tourniquet 5. 5 Applying CPR (Cardiopulmonary Resuscitation) 5. 6 Taking appropriate medicine

Curricular Content Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures. 1.2 Work safety procedures in a textile mill. 1.3 Emergency procedures. 1.3.1 Firefighting 1.3.2 Earthquake response. 1.3.3 Explosion response. 1.3.4 Accident response. 1.4 Types of hazards (biological, chemical, physical, ergonomic and electrical) and their effects.

	1.5 PPE types and uses. 1.6 Personal hygiene practices. 1.7 Safety signs, symbols and colour codes. 1.8 OHS awareness and reporting requirements.
2. Underpinning Skills	2. 1 Identifying OHS policies and procedures 2. 2 Following personal work safety practices 2. 3 Selecting and using appropriate PPE 2. 4 Reporting hazards and risks 2. 5 Responding to emergency procedures 2. 6 Maintaining physical well-being in the workplace 2. 7 Performing basic firefighting using fire extinguishers 2. 8 Applying basic first aid procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 PPEs 4.3 Firefighting equipment 4.4 Emergency response manual 4.5 Safety signs and symbols 4.6 First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 followed OHS policies and procedures. 1.2 selected and used personal protective equipment (PPE). 1.3 reported incidents arising from hazards and risks to the authority. 1.4 implemented emergency response plans and procedures. 1.5 applied basic first aid procedure.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after

	completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **WORK IN A TEAM ENVIRONMENT**	**Nominal Duration:** 10 Hrs.	**Unit Code:** SICIP-TEX-QCYM-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to work in a team environment. It specifically includes the tasks of identifying team goals and processes, communicating and cooperating with team members, working as a team member and solving problems as a team member.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and processes	1.1 Team goals and collaborative decision-making processes are identified. 1.2 Roles and responsibilities of team members are identified. 1.3 Relationships within the team and with other workers are identified.
2. Communicate and cooperate with team members	2. 1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives. 2. 2 Formal and informal **forms of communication** are used effectively to support team achievement. 2. 3 Diversity in character is respected and valued in team functioning. 2. 4 Views and opinions of other team members are understood and valued. 2. 5 Workplace terminology is used correctly to assist communication.
3. Work as a team member	3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with the team. 3.2 Tasks are performed in accordance with organizational and team requirements, specifications, and workplace procedures. 3.3 Support to other team members is provided to ensure the team achieves its goals. 3.4 Agreed reporting lines are followed using standard operating procedures.
4. Solve problems as a team member	4.1 Current and potential problems faced by the team are identified. 4.2 A solution to the problem is identified. 4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.

Range of Variables:

Variable	Range (May include but not limited to)
1. Forms of communication	1. 1 Agenda 1. 2 Simple reports such as progress and incident reports 1. 3 Job sheets and shift handover log 1. 4 Operational manuals and SOPs 1. 5 Brochures and promotional material 1. 6 Visual and graphic materials 1. 7 Standards 1. 8 OSH information 1. 9 Signs and symbols

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Team goals and collaborative decision-making processes 1. 2 Roles and responsibilities of team members 1. 3 Relationships within the team and with other workers 1. 4 Effective interpersonal skills to interact with team members 1. 5 Effective formal and informal forms of communication 1. 6 Value of diversity in team functioning 1. 7 Correct use of workplace terminology 1. 8 Team's duties, responsibilities, authorities, objectives and task requirements 1. 9 Support mechanisms for other members of the team to ensure achievement of goals 1. 10 Methods of identifying current and potential problems faced by a team 1. 11 Effective problem-solving methods and evaluation of outcomes
2. Underpinning Skills	2. 1 Identifying team goals and collaborative decision-making processes 2. 2 Identifying roles and responsibilities of team members 2. 3 Using effective interpersonal skills to interact with team members 2. 4 Using formal and informal forms of communication 2. 5 Understanding and valuing views and opinions of other team members 2. 6 Performing tasks in accordance with organizational and team requirements, specifications and workplace procedures 2. 7 Supporting other members of the team to ensure the team achieves its goals 2. 8 Identifying current and potential problems faced by the team 2. 9 Identifying solutions to problems 2. 10 Solving problems effectively and evaluating the outcome of the implemented solution

3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Manuals/Guidelines 4.3 Pens 4.4 Papers 4.5 Workbooks

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 identified team goals and work processes. 1. 2 communicated and cooperated with team members. 1. 3 worked as a team member. 1. 4 solved problems as a team member.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 4. Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: **WORK IN THE TEXTILE SECTOR**	**Nominal Duration:** 10 Hrs.	**Unit Code:** SICIP-TEX-QCYM-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for work in the textile sector. It specifically includes the tasks of identifying the textile sector and its main activities, understanding basic roles in textile production processes, recognizing common tools and equipment used, and understanding simple quality requirements in textile work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify the textile sector and its main activities	1. 1 The textile sector is identified within the context of national and global industries. 1. 2 **Main activities** of the textile sector are recognized. 1. 3 **Different stages of the textile value chain** are described in simple terms. 1. 4 Key workplaces in the textile sector are identified. 1. 5 Importance of the textile sector in the economy and the environment is understood.
2. Understand basic roles in textile production processes	2. 1 **Basic job roles in textile production** are identified and described. 2. 2 Simple production workflows and task sequences are understood. 2. 3 Teamwork and coordination in production processes are identified. 2. 4 Workplace safety roles and responsibilities are understood and followed. 2. 5 Shift working practices and reporting lines are understood.
3. Recognize common tools and equipment used	3.1 **Common tools used in textile processing** are identified correctly. 3.2 **Basic machines and equipment used in production** are recognized. 3.3 Functions of tools and equipment are understood in simple terms. 3.4 Basic maintenance and cleaning needs of tools and equipment are recognized.
4. Understand simple quality requirements in textile work	4.1 Basic quality standards in textile work are identified. 4.2 **Simple defects in textile products** are recognized. 4.3 Importance of maintaining quality during production is understood. 4.4 Work output is checked against basic quality

Elements of Competency	Performance Criteria
	expectations. 4.5 Good workmanship practices are followed to ensure quality.

Range of Variables:

Variable	Range (May include but not limited to)
1. Main activities	1. 1 Fibre production and preparation 1. 2 Yarn manufacturing (spinning) 1. 3 Fabric manufacturing (weaving and knitting) 1. 4 Wet processing (dyeing, printing and finishing) 1. 5 Garment manufacturing 1. 6 Testing and quality assurance 1. 7 Packaging, storage and export
2. Different stages of the textile value chain	2. 1 Natural and man-made fibre production 2. 2 Fibre preparation and mixing 2. 3 Yarn manufacturing (spinning) 2. 4 Fabric manufacturing (weaving/knitting) 2. 5 Dyeing, printing and finishing 2. 6 Cutting, sewing and finishing of garments 2. 7 Packaging, distribution and export
3. Basic job roles in textile production	3.1 Blow room operator/helper 3.2 Card tenter 3.3 Draw frame and comber operator 3.4 Simplex (speed frame) operator 3.5 Ring frame sider and doffer 3.6 Winding and packing operator 3.7 Quality checker/laboratory assistant 3.8 Maintenance fitter/electrician 3.9 Line leader/supervisor
4. Common tools used in textile processing	4.1 Sliver and roving cutter, scissors and nippers 4.2 Hooks, brushes and dusting tools 4.3 Spanner, screwdriver and plier set 4.4 Oil can and grease gun 4.5 Feeler and setting gauges 4.6 Wrap reel, wrap block and electronic balance 4.7 Tachometer and hand lens 4.8 Vacuum cleaner and air blowing gun
5. Basic machines and equipment used in production	5. 1 Bale opener/plucker and blow room line 5. 2 Carding machine 5. 3 Draw frame (breaker and finisher) 5. 4 Lap former and comber 5. 5 Speed frame (simplex) 5. 6 Ring frame 5. 7 Winding machine/autoconer

Variable	Range (May include but not limited to)
	5. 8 Humidification and waste collection plant 5. 9 Testing laboratory equipment
6. Simple defects in textile products	6. 1 Neps and slubs 6. 2 Thick and thin places 6. 3 Count variation and uneven yarn 6. 4 High hairiness 6. 5 Foreign fibre and contamination 6. 6 Sliver and roving breakage 6. 7 Soft/hard and damaged packages 6. 8 Oil and dirt stains

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Structure and importance of the textile sector in the national and global economy. 1. 2 Main activities and stages of the textile value chain. 1. 3 Position of yarn manufacturing within the textile value chain. 1. 4 Basic job roles, responsibilities and reporting lines in textile production. 1. 5 Simple production workflows, task sequences and shift practices. 1. 6 Common tools, machines and equipment used in textile production and their functions. 1. 7 Basic maintenance and cleaning needs of tools and equipment. 1. 8 Basic quality standards and common defects in textile products. 1. 9 Importance of quality, productivity and good workmanship in the workplace.
2. Underpinning Skills	2. 1 Identifying the textile sector and its main activities 2. 2 Describing the stages of the textile value chain in simple terms 2. 3 Understanding basic job roles and responsibilities in textile production processes 2. 4 Following simple production workflows, task sequences and teamwork practices 2. 5 Applying workplace safety roles and responsibilities in production activities 2. 6 Recognizing common tools, machines and equipment and their functions 2. 7 Recognizing basic maintenance and cleaning needs of tools and equipment 2. 8 Recognizing simple defects and checking work output against basic quality requirements 2. 9 Following good workmanship practices to ensure

	consistent quality output
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Charts, models and samples of fibre, sliver, roving and yarn 4.3 Multimedia projector, laptop and audio-visual clips 4.4 Learning manual and handouts 4.5 Stationery

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 identified the textile sector and its main activities. 1. 2 understood basic roles in textile production processes. 1. 3 recognized common tools and equipment used. 1. 4 understood simple quality requirements in textile work.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **APPLY GREEN PRACTICES IN THE TEXTILE SECTOR**	**Nominal Duration:** 10 Hrs.	**Unit Code:** SICIP-TEX-QCYM-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply green practices in the textile sector. It specifically includes the tasks of interpreting green concepts, minimizing resource use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1. 1 Basic concepts of environmental sustainability and green practices are explained clearly. 1. 2 **Environmental impacts of textile production** processes are identified. 1. 3 **Resource efficiency concepts** are interpreted. 1. 4 Waste reduction, reuse and recycling principles are described in relation to yarn manufacturing operations. 1. 5 **Eco-friendly practices in the textile sector** are recognized and related to workplace activities.
2. Minimize resource use in the workplace	2. 1 Electricity, water, compressed air and raw materials are used efficiently according to workplace requirements. 2. 2 **Unnecessary consumption of resources** is identified and avoided during operations. 2. 3 Machines and equipment are operated in energy-efficient modes where applicable. 2. 4 Material wastage is minimized through careful handling and accurate measurement. 2. 5 **Resource-saving practices** are applied where possible. 2. 6 Leakage, spillage and wastage of materials are promptly controlled and reported. 2. 7 **Workplace practices** for resource conservation are followed.
3. Implement waste management practices	3.1 **Types of waste** generated in yarn manufacturing operations are identified correctly. 3.2 Waste materials are segregated according to workplace and environmental requirements. 3.3 **Reusable and recyclable materials** are separated from non-recyclable waste. 3.4 Waste is stored safely in designated containers or areas. 3.5 Waste disposal is carried out following environmental rules and organizational procedures.

Elements of Competency	Performance Criteria
	3.6 Hazardous waste is handled carefully according to safety guidelines. 3.7 Waste reduction practices are applied to minimize environmental impact during operations.

Range of Variables:

Variable	Range (May include but not limited to)
1. Environmental impacts of textile production	1. 1 High electrical energy consumption of spinning machinery and humidification plants 1. 2 Cotton dust, fibre fly and micro-fibre emission 1. 3 Noise pollution from machines and compressors 1. 4 Solid waste generation (droppings, comber noil, soft and hard waste) 1. 5 Water consumption and effluent from wet processing 1. 6 Air emission and greenhouse gases from boilers and generators 1. 7 Used oil, grease and contaminated cotton waste
2. Resource efficiency concepts	2. 1 Energy-efficient motors and inverter/variable frequency drives 2. 2 Control of compressed air leakage and optimized air pressure 2. 3 Optimized humidification and ventilation control 2. 4 Reduction of soft and hard waste percentage 2. 5 Preventive maintenance to reduce power consumption 2. 6 Energy-efficient lighting and switching-off practices 2. 7 Optimum machine loading and reduction of idle running
3. Eco-friendly practices in the textile sector	3.1 Recycling of soft waste back into mixing where permitted 3.2 Use of recycled, organic or sustainably sourced fibre 3.3 Dust collection, filter and waste recovery systems 3.4 Compliance with Department of Environment (DoE) rules and regulations 3.5 Effluent treatment plant (ETP) and rainwater harvesting 3.6 Reuse of cans, bobbins, cones, cartons and packaging materials 3.7 Green building and energy audit practices
4. Unnecessary consumption of resources	4.1 Idle running of machines and motors 4.2 Compressed air leakage in pipelines and guns 4.3 Excess lighting in unoccupied areas 4.4 Over-cleaning with air guns instead of vacuum 4.5 Excess lubricant and cleaning material use 4.6 Excessive good fibre loss with droppings and noil
5. Resource-saving practices	5. 1 Switching off machines, lights and fans when not required

Variable	Range (May include but not limited to)
	5. 2 Reporting air, water and oil leakage promptly 5. 3 Using vacuum cleaning instead of air blowing where possible 5. 4 Reusing cans, tubes, cones and packaging materials 5. 5 Maintaining correct machine settings to reduce waste 5. 6 Recording and monitoring waste and energy use
6. Workplace practices	6. 1 Standard operating procedures (SOPs) 6. 2 Environmental policy of the organization 6. 3 Housekeeping and 5S practices 6. 4 Energy and waste monitoring records 6. 5 Instructions from supervisors and safety notices
7. Types of waste	7.1 Blow room droppings and card droppings/flat strips 7.2 Comber noil 7.3 Sliver, roving and pneumafil soft waste 7.4 Hard waste from winding (yarn and thread waste) 7.5 Damaged cans, bobbins, cones and tubes 7.6 Packaging waste (poly bags, bale wrappers, straps, cartons) 7.7 Used oil, grease and oil-contaminated cotton (hazardous waste) 7.8 Used filters, sweeps and floor waste
8. Reusable and recyclable materials	8.1 Usable soft waste for back-mixing 8.2 Comber noil for coarse count and open-end spinning 8.3 Cans, bobbins, tubes and cones 8.4 Cartons, poly bags and bale straps 8.5 Metal and plastic scrap 8.6 Paper and stationery waste

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Basic concepts of environmental sustainability, green practices and climate-resilient production. 1. 2 Environmental impacts of textile and yarn manufacturing operations. 1. 3 Resource efficiency concepts relating to energy, water, compressed air and raw material. 1. 4 Principles of waste reduction, reuse and recycling in a spinning mill. 1. 5 Eco-friendly practices and technologies applied in the textile sector. 1. 6 Types of waste generated at each stage of yarn manufacturing and permissible waste levels. 1. 7 Waste segregation, storage, disposal methods and compliance with environmental regulations. 1. 8 Hazardous waste handling and workplace environmental safety procedures.

	1. 9 Relevant national environmental rules and organizational environmental policy.
2. Underpinning Skills	2. 1 Interpreting green concepts and relating them to workplace activities 2. 2 Identifying environmental impacts of yarn manufacturing operations 2. 3 Applying resource efficiency principles in workplace operations 2. 4 Using electricity, compressed air, water and raw materials efficiently 2. 5 Identifying and avoiding unnecessary consumption of resources 2. 6 Operating machines and equipment in energy-efficient modes where applicable 2. 7 Handling materials carefully to reduce wastage and improve accuracy 2. 8 Controlling and reporting leakage, spillage and resource wastage promptly 2. 9 Identifying and segregating different types of waste generated in yarn manufacturing 2. 10 Separating reusable and recyclable materials from non-recyclable waste 2. 11 Handling hazardous waste according to safety guidelines and regulations
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4. 1 Workplace (simulated or actual) 4. 2 Standard operating procedures 4. 3 Workplace documents, signs and symbols 4. 4 Codes of conduct and environmental policy 4. 5 Waste segregation bins and storage area 4. 6 Tools, equipment and facilities appropriate to the process or activities 4. 7 Materials relevant to the proposed activity

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 interpreted green concepts. 1. 2 minimized resource use in the workplace. 1. 3 implemented waste management practices.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: **INTERPRET QUALITY CONCEPTS IN YARN MANUFACTURING**	**Nominal Duration:** 50 Hrs.	**Unit Code:** SICIP-TEX-QCYM-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to interpret productivity and quality concepts in yarn manufacturing. It specifically includes the tasks of interpreting quality concepts and standards, interpreting yarn testing methods, identifying factors affecting quality control procedures, and interpreting quality management system..		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret quality concepts and standards	1.1 The concepts of quality, quality control and quality assurance are explained and differentiated. 1.2 **Yarn quality characteristics** and their effect on downstream processes are identified. 1.3 Quality standards and reference systems applicable to yarn are interpreted.
2. Interpret yarn testing methods	2.1 **Common yarn testing methods** are identified and explained according to standard operating procedures. 2.2 Testing instruments and equipment are identified and their functions interpreted. 2.3 **Testing standards** are interpreted according to workplace quality requirements. 2.4 Test results are read, compared with yarn quality specifications, and deviations are reported to the supervisor.
3. Identify factors affecting quality control procedures	3.1 **Factors affecting quality** at each process stage are identified. 3.2 Factors affecting quality at each process stage are explained. 3.3 Identified factors are documented and reported.
4. Interpret quality management system	4.1 **Quality management system (QMS)** principles and requirements are interpreted. 4.2 Roles, responsibilities and authorities for quality in the workplace are identified. 4.3 Customer and compliance requirements applicable to the mill are identified.

Range of Variables:

Variable	Range (May include but not limited to)
1. Yarn quality characteristics	1.1 Count and count variation (CV%) 1.2 Twist per inch/metre and twist variation 1.3 Strength – count strength product (CSP), tenacity (RKM/cN.tex^{-1}) and elongation 1.4 Evenness (U% and CV%) 1.5 Imperfections – thin places, thick places and neps per kilometre 1.6 Hairiness index and its variation 1.7 Classimat/objectionable faults 1.8 Contamination and foreign fibre level
2. Common yarn testing methods	2.1 Yarn count 2.2 Yarn twist 2.3 Testing of strength 2.4 Evenness of yarn 2.5 Moisture regains
3. Testing standards	3.1 International Organization for Standardization (ISO) 3.2 American Society for Testing and Material (ASTM) 3.3 American Association of Textile Chemists and Colorists (AATCC)
4. Factors affecting quality	4.1 Raw materials 4.2 Process 4.3 Workforce 4.4 Machine 4.5 Methods 4.6 Environment
5. Quality management system (QMS)	5.1 ISO 9001 quality management system requirements 5.2 Quality management principles – customer focus, leadership, engagement of people, process approach, improvement, evidence-based decision making, relationship management 5.3 Total Quality Management (TQM) concept 5.4 Process approach and the plan-do-check-act cycle 5.5 Risk-based thinking (awareness level)

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Concepts of quality, quality control (QC) and quality assurance (QA), and the differences between them 1.2 Factors affecting quality at each stage of the spinning process. 1.3 Yarn quality characteristics and their effects on downstream processes 1.4 Common yarn testing methods and the instruments used for them 1.5 Testing and quality standards and reference systems applicable to yarn 1.6 Factors affecting quality at each process stage 1.7 Basic technical calculations used in a spinning mill 1.8 Principles and requirements of the quality management system 1.9 Roles, responsibilities and authorities for quality in the workplace 1.10 Customer and compliance requirements applicable to the mill
2. Underpinning Skills	2.1 Interpreting SOPs, setting sheets, mixing plans and lot particulars 2.2 Explaining quality concepts in simple technical language 2.3 Differentiating between quality control and quality assurance 2.4 Identifying yarn quality characteristics and relating them to the performance of the yarn in later processes 2.5 Reading and interpreting test results and comparing them with yarn quality specifications 2.6 Using and interpreting testing standards according to workplace quality requirements 2.7 Carrying out basic technical calculations (count, twist, production, efficiency, waste and CV%) using a calculator 2.8 Identifying, analyzing and documenting factors affecting quality at each process stage 2.9 Interpreting QMS principles and applying them to workplace situations 2.10 Reporting deviations and quality problems clearly to the supervisor, orally and in writing
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource	The following resources must be provided:

Implications	4.1 Classroom and workplace (simulated or actual) 4.2 Multimedia projector and computer with internet access 4.3 Spinning mill or training spinning laboratory (actual or simulated) for demonstration 4.4 Yarn samples of different counts and types, with test reports and quality specifications 4.5 Yarn testing instruments and equipment 4.6 Copies or extracts of quality standards (ISO, ASTM, AATCC) and ISO 9001 documents 4.7 Quality records and formats 4.8 Calculator and calculation worksheets, with learning guides, manuals and handouts 4.9 Charts and posters showing yarn faults and quality parameters 4.10 Learning manual and reference books

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 interpreted quality concepts and standards. 1.2 interpreted yarn testing methods. 1.3 identified factors affecting quality control procedures. 1.4 interpreted quality management system.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM FIBRE AND RAW MATERIAL QUALITY ASSESSMENT**	**Nominal Duration:** 60 Hrs.	**Unit Code:** SICIP-TEX-QCYM-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform fibre and raw material quality assessment. It specifically includes the tasks of preparing for fibre and raw material testing, collecting and conditioning samples, testing fibre properties, interpreting fibre test results, and recording and reporting fibre quality data.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for fibre and raw material testing	1.1 The testing requirement is determined from the **lot and mixing plan** and work instruction. 1.2 **Fibre testing instruments** are selected and checked for readiness and calibration status. 1.3 Standard atmospheric condition of the laboratory is verified and maintained. 1.4 Test formats, reference standards and specifications are arranged. 1.5 PPE and laboratory safety requirements are applied.
2. Collect and condition samples	2.1 The **sampling procedure** for bales and mixing is followed correctly. 2.2 Representative samples are drawn from the specified locations and depths. 2.3 Samples are identified, labelled and made traceable to the bale and lot. 2.4 Samples are conditioned for the required period before testing. 2.5 Samples are handled and preserved to avoid contamination, damage and moisture change.
3. Test fibre properties	3.1 **Fibre properties** are tested following the applicable **test methods**. 3.2 Instruments are operated as per the operating procedure and manufacturer's manual. 3.3 The required number of readings is taken and abnormal readings are repeated. 3.4 Trash and contamination content is determined. 3.5 Moisture content and regain of the raw material are determined.
4. Interpret fibre test results	4.1 Test results are compared with **specification and standard values**. 4.2 The effect of fibre properties on yarn quality and spinning performance is explained.

Elements of Competency	Performance Criteria
	4.3 Bale grouping and mixing recommendations are provided to the supervisor.
5. Record and report fibre quality data	5.1 Test data are recorded in the **prescribed formats** accurately and legibly. 5.2 Bale-wise and lot-wise summaries, averages and CV% are prepared. 5.3 Records are filed and preserved for traceability and audit. 5.4 Instruments, samples and the work area are cleaned and left in order.

Range of Variables:

Variable	Range (May include but not limited to)
1. Lot and mixing plan	1.1 Bale laydown plan and bale grouping 1.2 Mixing percentage of different origins and grades 1.3 Blend ratio (cotton/polyester/viscose and other blends) 1.4 Target count, quality level and end use of the lot
2. Fibre testing instruments	2.1 High Volume Instrument (HVI) system 2.2 Advanced Fibre Information System (AFIS) or equivalent 2.3 Micronaire (fineness) tester 2.4 Fibrograph, comb sorter or Baer sorter for length measurement 2.5 Pressley/Stelometer or bundle strength tester 2.6 Electronic balance (0.001 g accuracy)
3. Sampling procedure	3.1 Percentage and number of bales sampled per lot 3.2 Sampling location and depth within a bale 3.3 Number, size and mass of samples 3.4 Blending and reduction of the gross sample to the test sample 3.5 Sample identification, labelling and traceability 3.6 Conditioning at 20 ± 2 °C and 65 ± 2% RH 3.7 Minimum conditioning time before testing
4. Fibre properties	4.1 Length – upper half mean length (UHML), mean length, uniformity index, short fibre index 4.2 Fineness – micronaire value, denier/decitex for man-made fibre 4.3 Strength (g/tex) and elongation percentage 4.4 Colour (Rd, +b), grade and appearance 4.5 Trash and leaf content 4.6 Neps per gram (seed coat and fibre neps) 4.7 Moisture content and regain 4.8 Staple length, crimp and lustre of man-made fibre 4.9 Spinning Consistency Index (SCI)
5. Test methods	5.1 HVI and AFIS operating procedures 5.2 Manual length measurement (comb sorter/Baer sorter)

Variable	Range (May include but not limited to)
	method 5.3 Manual nep counting and appearance assessment
6. Specification and standard values	6.1 HVI/USTER reference values for the origin and grade 6.2 Fibre quality index and spinning consistency index norms
7. Prescribed formats	7.1 Bale-wise test record 7.2 Lot summary and mixing quality sheet 7.3 Bale laydown and mixing record 7.4 Trash, moisture and contamination report 7.5 Laboratory register and instrument log book

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Types, sources and classification of textile fibres used in spinning. 1.2 Physical and chemical properties of cotton and man-made fibres. 1.3 Importance of raw material quality in yarn productivity, quality and cost. 1.4 Bale management, laydown and mixing principles. 1.5 Principles of sampling and representative sample collection. 1.6 Standard atmospheric conditions and the need for conditioning. 1.7 Fibre testing instruments – working principle, operation, care and calibration. 1.8 Standard test methods for length, fineness, strength, maturity, trash and moisture. 1.9 Relationship between fibre properties and yarn strength, evenness, imperfections and hairiness. 1.10 Effect of raw material on end breakage rate, waste percentage and yarn realization.
2. Underpinning Skills	2.1 Preparing the laboratory and instruments for fibre testing 2.2 Drawing representative samples from bales and mixing 2.3 Labelling, conditioning and preserving samples 2.4 Operating HVI/AFIS and manual fibre testing instruments 2.5 Determining fibre length, fineness, strength, maturity and colour 2.6 Determining trash content, nep content and moisture regain 2.7 Interpreting fibre test reports for bale grouping and mixing decisions 2.8 Recording, summarizing and reporting fibre quality data 2.9 Maintaining cleanliness, care and safe handling of testing instruments
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties

	3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Testing laboratory with standard atmospheric control 4.2 HVI/AFIS or equivalent fibre testing system 4.3 Shirley trash analyser 4.4 Moisture meter and hot air oven 4.5 Electronic balance (0.001 g accuracy) 4.6 Fibre samples of different origins, grades and types 4.7 Bale sampling tools, sample bags and labels 4.8 PPE and laboratory safety equipment 4.9 Learning manual and audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 prepared for fibre and raw material testing. 1.2 collected and conditioned representative samples. 1.3 tested fibre properties following the applicable test methods. 1.4 interpreted fibre test results against specification and standard values. 1.5 recorded and reported fibre quality data.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM IN-PROCESS QUALITY CONTROL IN SPINNING PREPARATORY PROCESSES**	**Nominal Duration:** 90 Hrs.	**Unit Code:** SICIP-TEX-QCYM-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform in-process quality control in spinning preparatory processes. It specifically includes the tasks of preparing for in-process quality control, checking blow room, lap and chute feed quality, checking card sliver quality, checking draw frame, comber and roving quality, and recording and reporting in-process quality data.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for in-process quality control	1.1 The control plan and testing schedule are interpreted correctly. 1.2 **In-process testing instruments** and accessories are arranged and checked for readiness. 1.3 Process stages, machines and sampling points are identified. 1.4 Standard values and **tolerance limits** for each stage are obtained from workplace documents. 1.5 Safety procedures for working on and near running machines are followed.
2. Check blow room, lap and chute feed quality	2.1 Samples are collected from the blow room delivery as per the schedule. 2.2 Lap weight, chute feed weight and their variation are checked against the standard. 2.3 **Trash and cleaning efficiency** of the blow room line is determined. 2.4 Nep increase and indications of fibre damage are checked.
3. Check card sliver quality	3.1 Card sliver hank and weight per unit length are checked at the specified frequency. 3.2 Sliver evenness (A%, U%, CV%) is checked using the applicable instrument. 3.3 **Card quality parameters** are checked against the standard. 3.4 Cleaning efficiency and waste percentage of the card are determined. 3.5 Card-to-card and delivery-to-delivery variation is compared. 3.6 Out-of-limit results are reported and corrective action is followed up.
4. Check draw frame, comber and roving	4.1 Sliver hank and weight variation at breaker and finisher drawing are checked. 4.2 **Sliver evenness and spectrogram** results are examined

Elements of Competency	Performance Criteria
quality	for periodic and random faults. 4.3 Auto-leveller performance is verified against the standard. 4.4 Comber lap quality, noil percentage and combed sliver quality are checked. 4.5 Roving hank, twist multiplier and unevenness are checked against the standard. 4.6 Roving bobbin build, tension and breakage rate are observed and recorded.
5. Record and report in-process quality data	5.1 All test results are recorded in the **prescribed formats**. 5.2 Machine-wise and shift-wise summaries are prepared correctly. 5.3 Reports are submitted to the authority within the required time. 5.4 Non-conformances are escalated through the correct reporting channel. 5.5 Records are maintained legibly, completely and traceably.

Range of Variables:

Variable	Range (May include but not limited to)
1. In-process testing instruments	1.1 Wrap reel and wrap block for sliver, roving and yarn 1.2 Digital lap and sliver weighing scale 1.3 Electronic balance (0.01 g and 0.001 g accuracy) 1.4 Evenness tester (capacitive or optical) with spectrogram facility 1.5 Trash analyser 1.6 Nep counting board and hand lens 1.7 Twist tester 1.8 Stop watch and tachometer
2. Tolerance limits	2.1 Hank/count variation limits at each stage 2.2 A%, U% and CV% limits for lap, sliver and roving 2.3 CVb% between deliveries and between machines 2.4 Trash percentage and cleaning efficiency limits 2.5 Nep limit per gram at card and comber 2.6 Waste and noil percentage range 2.7 Twist multiplier range for roving 2.8 Breakage rate limit at each stage
3. Trash and cleaning efficiency	3.1 Trash percentage in mixing and in blow room delivery 3.2 Cleaning efficiency percentage of each machine and of the line 3.3 Waste percentage of each machine 3.4 Trash particle size distribution 3.5 Nep generation across the line
4. Card quality parameters	4.1 Sliver hank and hank variation 4.2 Neps per gram in card sliver

Variable	Range (May include but not limited to)
	4.3 Trash percentage in card sliver 4.4 Flat strip, licker-in and cylinder waste percentage 4.5 Cleaning efficiency of the card 4.6 Short fibre content in card sliver 4.7 Sliver A%, U% and CV% 4.8 Card clothing condition and wire sharpness
5. Sliver evenness and spectrogram	5.1 U% and CV% of sliver and roving 5.2 Spectrogram – mechanical waves and drafting waves 5.3 A% (deviation from nominal hank) 5.4 Thick and thin places in sliver 5.5 Count/hank CV over 1 m, 3 m, 10 m and 50 m
6. Prescribed formats	6.1 Hank/count check sheet 6.2 Evenness and spectrogram test report 6.3 Trash, waste and cleaning efficiency report 6.4 Noil percentage and comber performance record 6.5 Roving quality and breakage record 6.6 Control chart sheet 6.7 Shift quality summary and handover log

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Objectives and principles of in-process quality control in a spinning mill. 1.2 Sequence of preparatory processes and the quality function of each machine. 1.3 Sampling techniques and sampling points for lap, sliver and roving. 1.4 Working principle, operation and care of in-process testing instruments. 1.5 Methods of determining hank, weight variation, evenness and CV%. 1.6 Trash analysis, cleaning efficiency and lint loss calculation. 1.7 Nep formation, nep counting and factors affecting nep level. 1.8 Comber performance parameters – noil percentage, lap quality and combed sliver quality. 1.9 Roving quality parameters – hank, twist multiplier, unevenness and bobbin build. 1.10 Common process faults, their probable causes and corrective actions.
2. Underpinning Skills	2.1 Interpreting the control plan and testing schedule 2.2 Selecting, checking and using in-process testing instruments 2.3 Collecting representative samples of lap, sliver and roving

	2.4 Measuring hank, weight per unit length and weight variation 2.5 Operating the evenness tester and obtaining U%, CV%, A% and spectrogram 2.6 Determining trash percentage, cleaning efficiency and lint loss 2.7 Counting neps and assessing sliver appearance 2.8 Determining comber noil percentage and lap quality 2.9 Checking roving hank, twist and bobbin build 2.10 Recording, summarizing and reporting in-process quality data
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) with preparatory machinery 4.2 Blow room line, carding machine, draw frame, lap former and comber 4.3 Speed frame (simplex) 4.4 Testing laboratory with standard atmospheric control 4.5 Wrap reel, wrap block and digital sliver/lap weighing scale 4.6 Electronic balance (0.001 g and 0.01 g accuracy) 4.7 Evenness tester with spectrogram facility 4.8 Twist tester, tachometer and stop watch 4.9 Hygrometer, thermometer and lux meter 4.10 Sample lap, sliver and roving of different counts 4.11 Control plan, SOPs, standards and recording formats 4.12 PPE and learning manual with audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 prepared for in-process quality control. 1.2 checked blow room, lap and chute feed quality. 1.3 checked card sliver quality. 1.4 checked draw frame, comber and roving quality. 1.5 recorded and reported in-process quality data.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test

	2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM YARN QUALITY TESTING AND EVALUATION**	**Nominal Duration:** 70 Hrs.	**Unit Code:** SICIP-TEX-QCYM-04-O

Unit Descriptor:
This unit covers the skills, knowledge and attitudes required to perform yarn quality testing and evaluation. It specifically includes the tasks of preparing the laboratory and instruments for yarn testing, testing yarn count, twist and moisture regain, testing yarn strength and elongation, testing yarn evenness, imperfections and hairiness, and interpreting, recording and reporting yarn test results.

Elements and Performance Criteria:
(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare the laboratory and instruments for yarn testing	1.1 **Yarn testing instruments** are selected, cleaned and checked for readiness. 1.2 Instrument calibration status and validity are verified before testing. 1.3 Standard atmospheric condition (20 ± 2 °C and 65 ± 2% RH) is verified and maintained. 1.4 Samples are drawn as per the **sampling procedure** and conditioned for the required period. 1.5 Laboratory safety and housekeeping requirements are followed.
2. Test yarn count, twist and moisture regain	2.1 Yarn count is determined using the wrap reel and electronic balance. 2.2 Count CV% and deviation from nominal count are calculated. 2.3 Twist per inch/metre is determined following the twist testing method. 2.4 Twist multiplier and twist CV% are calculated. 2.5 Moisture regain is determined and count is corrected to standard regain.
3. Test yarn strength and elongation	3.1 Lea strength and count strength product (CSP) are determined. 3.2 Single yarn strength, tenacity (RKM/cN.tex^{-1}) and elongation are determined. 3.3 Mean, CV%, minimum value and breaking elongation are calculated. 3.4 Results are compared with the standard and the customer requirement.
4. Test yarn evenness, imperfections and hairiness	4.1 Yarn evenness (U% and CV%) is measured using the evenness tester. 4.2 **Imperfections** are measured at the specified sensitivity levels. 4.3 Hairiness index (H) and its standard deviation are

Elements of Competency	Performance Criteria
	determined. 4.4 **Diagram, spectrogram and variance-length curve** are obtained and examined. 4.5 Test conditions (speed, test length and number of tests) are set as per the standard.
5. Interpret, record and report yarn test results	6.1 Test results are compared with **reference standards** and the customer specification. 6.2 Deviations and non-conformities are identified and reported promptly. 6.3 Results are recorded in the **prescribed formats** and plotted on control charts.

Range of Variables:

Variable	Range (May include but not limited to)
1. Yarn testing instruments	1.1 Wrap reel and wrap block 1.2 Electronic balance (0.001 g accuracy) 1.3 Twist tester (direct counting and untwist-retwist type) 1.4 Lea strength tester (CSP) 1.5 Single yarn strength tester (CRE type) with elongation measurement 1.6 Evenness tester with imperfection and spectrogram facility 1.7 Hairiness tester or hairiness module 1.8 Yarn fault classification system (Classimat or equivalent) 1.9 Black board winder and appearance grading board/cabinet 1.10 Moisture meter and hot air oven 1.11 Package and cone weighing scale
2. Sampling procedure	2.1 Number of packages/cones sampled per lot and per machine 2.2 Number of tests per package and total number of tests 2.3 Sample identification, labelling and traceability 2.4 Conditioning time and standard testing atmosphere
3. Imperfections	3.1 Thin places at −30%, −40% and −50% sensitivity 3.2 Thick places at +35% and +50% sensitivity 3.3 Neps at +140%, +200% and +280% sensitivity 3.4 Total imperfection index (IPI) per kilometre
4. Diagram, spectrogram and variance-length curve	4.1 Mass variation diagram at different cut lengths 4.2 Spectrogram showing mechanical and drafting waves 4.3 CV% at 1 m, 3 m, 10 m, 50 m and 100 m cut lengths
5. Reference standards	6.1 USTER STATISTICS percentile values (5%, 25%, 50%, 75%, 95%) 6.2 ASTM, ISO and BS standard test methods 6.3 Buyer specification, contract and end-use requirement

Variable	Range (May include but not limited to)
	6.4 Mill in-house norms and control limits 6.5 Previous lot results and trend data
6. Prescribed formats	7.1 Count, twist and strength test report 7.2 Evenness, imperfection and hairiness test report 7.3 Classimat/fault classification report 7.4 Lot-wise quality summary and quality certificate 7.5 Control chart sheet and trend char 7.6 Laboratory register and instrument log book

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Objectives and importance of yarn quality testing and evaluation. 1.2 Standard atmospheric conditions, conditioning and their effect on test results. 1.3 Sampling principles and sample size determination for yarn testing. 1.4 Working principle, operation, care and calibration of yarn testing instruments. 1.5 Standard test methods for count, twist, strength, elongation, evenness and hairiness. 1.6 Concept of evenness, index of irregularity, imperfections and hairiness. 1.7 USTER STATISTICS and the use of percentile values for benchmarking. 1.8 Effect of yarn quality characteristics on weaving, knitting and dyeing performance. 1.9 Preparation of test reports, quality certificates and documentation requirements.
2. Underpinning Skills	2.1 Preparing the laboratory and verifying atmospheric conditions 2.2 Verifying instrument calibration status before testing 2.3 Drawing, labelling and conditioning yarn samples 2.4 Determining yarn count using wrap reel and electronic balance 2.5 Determining twist per inch and twist multiplier 2.6 Determining lea strength, CSP, single yarn strength, tenacity and elongation 2.7 Operating the evenness tester and obtaining U%, CV% and imperfections 2.8 Comparing results with USTER STATISTICS and customer specification 2.9 Calculating mean, standard deviation, CV% and confidence limits 2.10 Preparing test reports
3. Underpinning	3.1 Commitment to occupational health and safety

Attitudes	3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Yarn testing laboratory with standard atmospheric control 4.2 Wrap reel, wrap block and electronic balance (0.001 g accuracy) 4.3 Twist tester 4.4 Lea strength tester and single yarn strength tester (CRE) 4.5 Evenness tester with imperfection, spectrogram and hairiness facility 4.6 Black board winder and appearance grading cabinet with standard photographs 4.7 Moisture meter 4.8 Conditioning chamber 4.9 Yarn samples of different counts, blends and quality levels 4.10 Standard test methods, specification sheets and recording formats 4.11 Computer with spreadsheet and reporting facility 4.12 PPE and learning manual with audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 prepared the laboratory and instruments for yarn testing. 1.2 tested yarn count, twist and moisture regain. 1.3 tested yarn strength and elongation. 1.4 tested yarn evenness, imperfections and hairiness. 1.5 interpreted, recorded and reported yarn test results.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency:	**Nominal Duration:**	**Unit Code:**

IDENTIFY AND CONTROL YARN FAULTS AND CONTAMINATION	50 Hrs.	SICIP-TEX-QCYM-05-O

Unit Descriptor:
This unit covers the skills, knowledge and attitudes required to identify and control yarn faults and contamination. It specifically includes the tasks of identifying yarn, package and process faults, analysing causes of faults, controlling contamination and foreign fibre, and applying corrective and preventive actions.

Elements and Performance Criteria:
(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify yarn, package and process faults	1.1 **Yarn faults** are identified visually and from test data. 1.2 **Package faults** are identified during winding, conditioning and packing. 1.3 Fault frequency, severity and distribution are assessed and classified. 1.4 Faulty material is segregated, marked and held as per workplace procedure.
2. Analyse causes of faults	2.1 Fault samples are examined and traced to the **originating process or machine**. 2.2 Machine, material, method, manpower and environment causes are separated. 2.3 Spindle-wise, machine-wise and shift-wise data are compared to locate the source.
3. Control contamination and foreign fibre	3.1 **Sources of contamination** are identified at each stage of the process. 3.2 **Contamination control measures** are implemented in the work area. 3.3 Contamination sorting and removal practices are monitored and verified. 3.4 Performance of the foreign fibre detector and yarn clearer is monitored.
4. Apply corrective and preventive actions	4.1 **Corrective actions** are recommended and implemented within the level of authority. 4.2 Preventive measures are proposed to avoid recurrence of the fault. 4.3 Machine settings, accessories and schedules are corrected through the authorized channel. 4.4 Affected material is reprocessed, downgraded or rejected as per procedure. 4.5 Operators are briefed on the correct work method where the cause is human.

Range of Variables:

Variable	Range (May include but not limited to)
1. Yarn faults	1.1 Thick places, thin places and neps 1.2 Slubs and spun-in fly 1.3 Count variation and mixed count 1.4 High and uneven hairiness 1.5 Foreign fibre and polypropylene contamination 1.6 Oil, grease and dirt stains 1.7 Double ends and multiple ends 1.8 Hard twist, soft twist and untwisted places 1.9 Barré-causing variation between machines and lots 1.10 Improper splice and bad knot
2. Package faults	2.1 Soft package and hard package 2.2 Bulged, dished, conical and saddle-shaped package 2.3 Damaged nose, base and cone body 2.4 Wrong package weight and length 2.5 Mixed count or mixed lot in a carton 2.6 Loose, missing or incorrect labelling 2.7 Improper conditioning of the package
3. Originating process or machine	3.1 Mixing and blow room 3.2 Carding 3.3 Breaker and finisher drawing 3.4 Lap former and comber 3.5 Speed frame (simplex) 3.6 Ring frame 3.7 Winding and yarn clearing 3.8 Conditioning, packing and dispatch 3.9 Humidification and utility plant 3.10 Material handling, storage and transport
4. Sources of contamination	4.1 Raw cotton, bale wrappers and bale straps 4.2 Packing materials (jute, polypropylene, plastic, paper) 4.3 Operator clothing, hair, buttons and personal items 4.4 Cleaning cloth, mops, brooms and dusters 4.5 Residue of the previous lot in machines, cans and trolleys 4.6 Oil, grease, rust and paint flakes 4.7 Insects, dust, fly and airborne fibre 4.8 Storage, transport and handling areas 4.9 Waste re-mixing and floor sweepings
5. Contamination control measures	5.1 Manual contamination sorting at the pre-opening/mixing stage 5.2 Use of cotton-only cleaning cloth and lint-free wipers 5.3 Controlled work clothing, hair covering and prohibition of loose items 5.4 Colour-coded and dedicated cans, trolleys and containers 5.5 Machine and can cleaning between lots and counts 5.6 Contamination/foreign fibre detector at blow room and winding

Variable	Range (May include but not limited to)
	5.7 Operator awareness, training and reward system 5.8 Regular contamination audit and monitoring 5.9 Segregation and controlled reuse of waste
6. Corrective actions	6.1 Correcting machine settings, gauges and drafting parameters 6.2 Replacing worn cots, aprons, travellers, rings, spindle tapes and card clothing 6.3 Adjusting yarn clearer settings and splicer parameters 6.4 Adjusting draft, twist and speed within permissible limits 6.5 Correcting temperature, relative humidity and air movement 6.6 Retraining operators on piecing, doffing and handling methods 6.7 Revising cleaning, lubrication and maintenance schedules 6.8 Segregating, reprocessing or downgrading defective material 6.9 Improving material handling, storage and identification practices

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Classification of yarn faults – frequent, seldom-occurring and objectionable. 1.2 Appearance, causes and effects of each type of yarn and package fault. 1.3 Fault tracing methods and identification of the originating machine. 1.4 Contamination control systems, detectors and clearing devices. 1.5 Yarn clearer types, clearing curve and setting principles. 1.6 Corrective and preventive action procedures and documentation. 1.7 Reporting, escalation and standardization requirements.
2. Underpinning Skills	2.1 Identifying yarn faults visually and from test data 2.2 Identifying package faults during winding and packing 2.3 Segregating, marking and preserving faulty material and samples 2.4 Tracing faults to the originating process, machine or spindle 2.5 Implementing and monitoring contamination control measures 2.6 Monitoring foreign fibre detector and yarn clearer performance 2.7 Measuring and reporting the contamination level 2.8 Recommending and implementing corrective and preventive actions

	2.9 Preparing fault analysis
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) with ring frame and winding machine 4.2 Yarn fault sample board and reference fault samples 4.3 Contamination and foreign fibre samples 4.4 Black board winder and appearance grading cabinet 4.5 Evenness tester with spectrogram facility 4.6 Yarn fault classification system (Classimat or equivalent) 4.7 Yarn clearer and splicer with setting facility 4.8 Hand lens, magnifier and inspection light 4.9 Colour-coded containers, trolleys and contamination sorting table 4.10 Computer with spreadsheet and charting facility 4.11 Learning manual with audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 identified yarn, package and process faults. 1.2 analysed causes of faults using appropriate techniques. 1.3 controlled contamination and foreign fibre in the work area. 1.4 applied corrective and preventive actions.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

-END OF COMPETENCY STANDARD-

WORKSHOP / LAB FACILITY STANDARD

Course Name:	Quality Control in Yarn Manufacturing
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/Lab):
1. Classroom – 500 sft (56 sqm)
2. Testing laboratory/workshop – 1500 sft (140 sqm),

Major Training Equipment and Training Facilities:

S.N.	Major Equipment and Training Facilities	Required Quantity
1.	Computer/Laptop	1
2.	Multimedia projector & screen	1
3.	Whiteboard with accessories	1
4.	High Volume Instrument (HVI) or equivalent fibre testing system	1
5.	Advanced Fiber Information System (AFIS) or equivalent	1
6.	Bundle strength tester (Pressley/Stelometer)	1
7.	Roving/Sliver/Yarn Testing Machine (USTER-4/USTER-6 or Equivalent)	1
8.	Moisture meter	1
9.	Wrap reel with wrap block (sliver, roving and yarn)	3
10.	Twist tester	2
11.	Lea strength tester (CSP)	1
12.	Single yarn strength tester (CRE type) with elongation measurement	1
13.	Yarn fault classification system (Classimat or equivalent)	1
14.	Tachometer/Stroboscope	1

The following conditions must be fulfilled –
1. The institute shall not use the same facilities for any other projects/organizations offering a similar course.
2. The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facilities required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirements of the Competency Standard (CS) provided by SICIP.

For the operation of the training course on Productivity Improvement and Quality Control in Yarn, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is

below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

This occupation requires observation, measurement and analysis to be carried out on running production machinery. Where a training institute does not possess a full-scale spinning line and a fully equipped yarn testing laboratory, the institute must establish a formal linkage with a spinning mill so that the practical sessions of the occupation-specific units – particularly in-process quality control, yarn testing, fault and contamination control, productivity measurement and waste control – can be conducted in an actual workplace under supervision.

The institute will also provide all other hand tools, instruments and consumables as per the CS for 25 trainees. The institute will also arrange adequate seating arrangements and classroom setup for the 25 trainees.